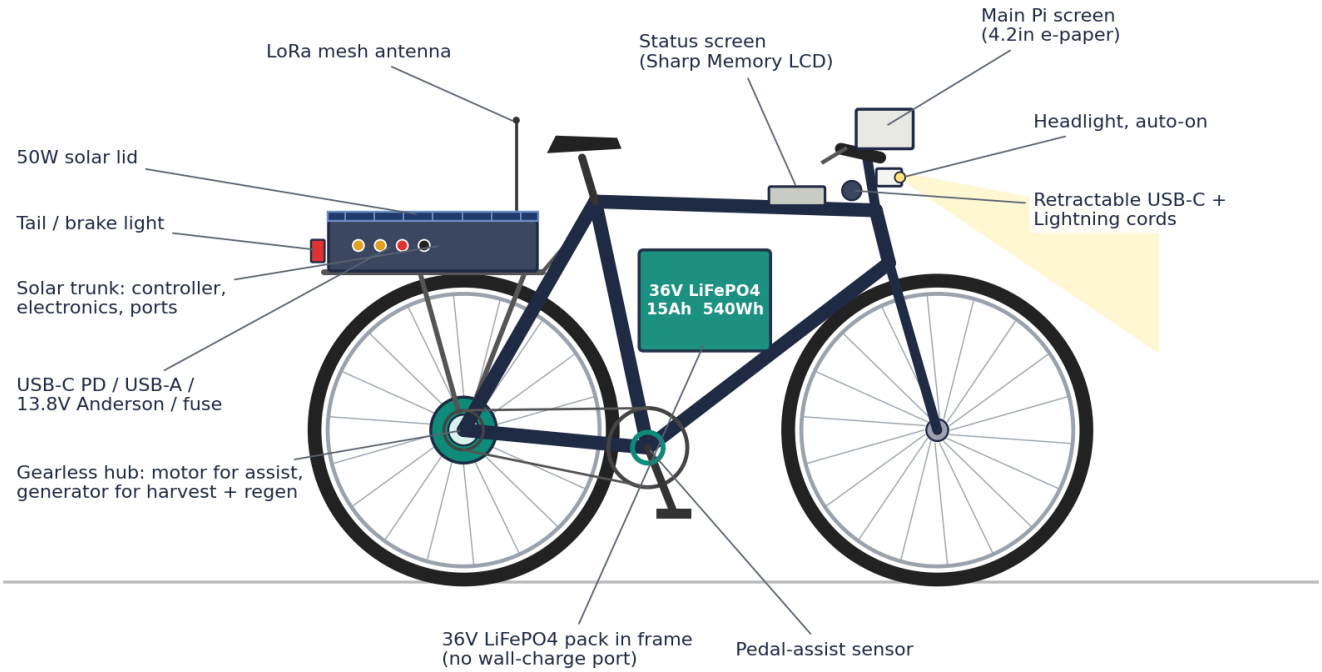
CONCEPT SPEC v0.2

# Pedal Power Station

An off-grid e-bike that never plugs into the wall. Ride it, park it on a stand, or let the sun work: every watt it uses is one you or the sun made. Pedal-assist up to 15 mph, solar backup, a headlight, built-in USB-C and Lightning cords, a 13.8V radio/car port, and a no-backlight Raspberry Pi command screen for emergency comms.

## Ride mode

Harvest, neutral or e-assist up to 15 mph, powered only by energy you made




| At a glance | |
|---|---|
| **Storage** | 36V LiFePO4 15Ah (about 540 Wh), built-in BMS. No wall-charge port. |
| **Charged by** | Pedaling (riding or on the stand), regen braking, and a 50W solar lid. Nothing else. |
| **E-assist** | 500W gearless hub, pedal-assist only, capped at 15 mph; about 10 Wh per mile |
| **Lights** | Auto-on headlight plus tail/brake light, powered from the pack |
| **Device charging** | Retractable USB-C and Lightning cords, USB-C PD and USB-A ports (5.3V) |
| **Radio / car** | Regulated 13.8V 10A Anderson port and car-battery clamp lead |
| **Screens + brains** | Always-on Sharp Memory LCD (Pico 2) + 4.2in e-paper (Pi Zero 2 W) |
| **Comms** | Meshtastic LoRa mesh node for off-grid text messaging |
| **Est. build cost** | about $1,250-2,330 including a used donor bike (see parts list) |

# 1. Your goals, and how the design meets them

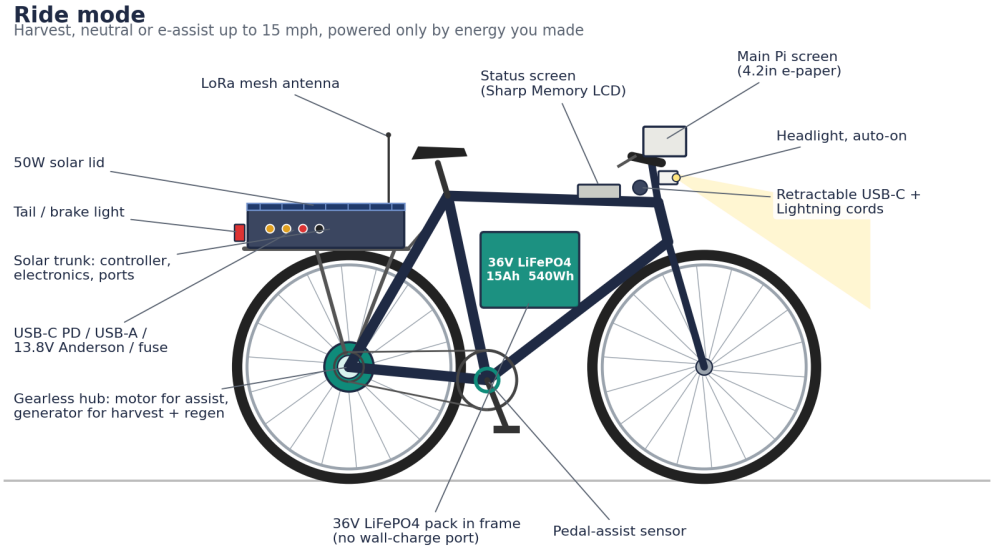
Every goal from the brief is covered. Two are met in a different place than first imagined, because that is where they work best on a real bike.

| Goal | How it is met | Note |
|---|---|---|
| **E-bike that never plugs in** | No charge port at all. The pack fills only from pedaling, regen braking and solar. | The assist you spend is assist you earned. |
| **15 mph top assisted speed** | Set as a hard limit in the motor controller; assist fades out above 15 mph. | You can still pedal faster unassisted. |
| **Ride or stay stationary** | Same bike. Fold-down stand and front riser block turn it into a porch generator in a minute. | One hub is motor and generator. |
| **Headlight for pedaling in the dark** | Auto-on headlight and tail/brake light, on in both ride and station mode. | About 1% of the pack per hour. |
| **Lightning + USB-C cords** | Retractable cords at the handlebar, plus USB-C PD and USB-A ports on the trunk. | Phones, tablets, power banks. |
| **12V / car battery charging** | Regulated 13.8V Anderson port plus a clamp lead with reverse-polarity protection. | Tops up a weak battery; not a jump-starter. |
| **Pi with no-backlight screen** | Pi Zero 2 W drives a 4.2in e-paper screen, readable in sun. | Pi sleeps when not needed. |
| **Secondary status screen** | Sharp Memory LCD run by a Pico: speed or watts, battery, assist left, time, solar. | Works even when the Pi is off. |
| **Flywheel for inertia** | Moved to the stationary stand (trainer roller with flywheel). | Heavy wheels would kill the e-bike range. |
| **Solar while out pedaling** | Moved from the wheels to a 50W lid on the rear trunk, with fold-out wings when parked. | Wheel cells get little area and constant shading. |
| **Emergency comms** | Meshtastic LoRa node; 13.8V port runs a mobile radio. | Texts with no cell service. |

**Why the flywheel and solar moved.** A flywheel stores energy only when it changes speed, so in a road wheel it mostly adds weight the motor and your legs must haul up every hill. On a stand it gives the smooth, coasting feel of a spin bike. Solar cells on spinning wheels see the sun at a constantly changing angle, are shaded by frame and spokes, and take road impacts; a flat lid panel makes several times more energy for less money.

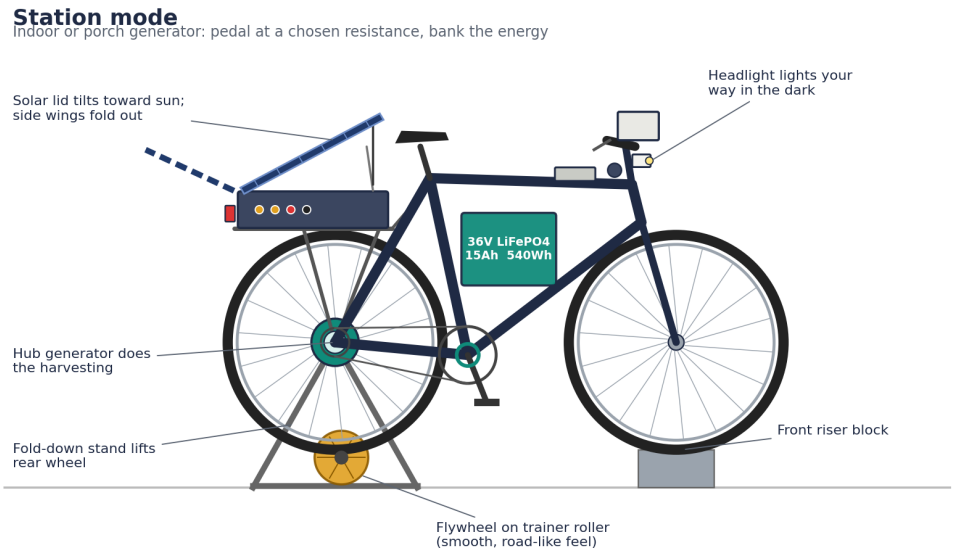
# 2. Two places to use it

**Out riding.** Choose Harvest, Neutral or Assist from the status screen (details in section 3). The solar lid charges the whole time you are outdoors, and the headlight turns on by itself at dusk.




*Ride mode: the heavy pack sits low in the frame triangle; electronics and ports live in the solar trunk.*

**On the stand.** Flip down the stand, set the front wheel on the riser block, and pedal at a chosen resistance level (L1-L5). The trainer roller and flywheel give road-like momentum while the hub generator banks the energy. The headlight doubles as a work light if you are pedaling during a blackout.




*Station mode: a used wheel-on trainer already has a roller, flywheel and clamp, so it can be adapted rather than built.*

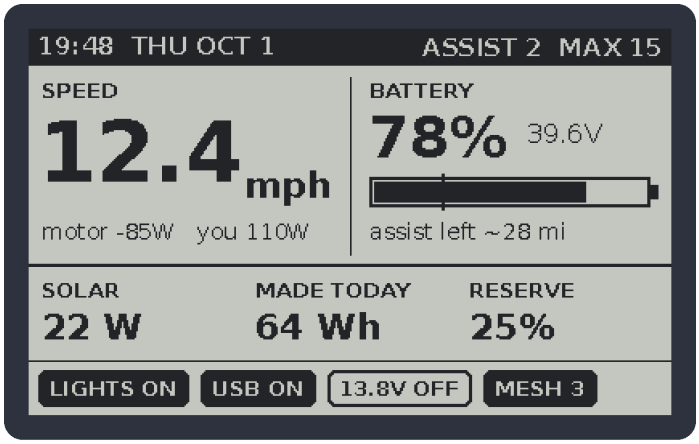
# 3. The e-bike that only runs on what you make

The same gearless hub that generates power can also push you along. A regen-capable motor controller runs it both ways, so switching between earning and spending energy is just a button press.

| Setting | What the motor does | Feels like |
|---|---|---|
| **Harvest (H1-H3)** | Acts as a generator and puts a light load on the wheel; energy flows into the pack. | Riding into a gentle breeze. Good on flats and downhills. |
| **Neutral** | Spins freely, no drag and no help. | A normal (slightly heavy) bike. |
| **Assist (A1-A3)** | Adds power in proportion to your pedaling, up to 15 mph. Pedal-assist only, no throttle. | A tailwind. Best for hills and headwinds. |
| **Braking** | Squeezing either brake lever cuts assist and switches on regen for a little energy back. | Slightly firmer rear braking. |

## Rules built into the firmware

- **15 mph cap.** Assist fades to zero between 14 and 15 mph. Set in the controller and checked again by the Pico.
- **Emergency reserve.** Assist shuts off when the pack falls to 25%. That last 25% (about 135 Wh) is kept for lights, phones, radios and the mesh node, so the bike stays useful in an outage. You can always pedal home.
- **Assist budget on screen.** The status screen turns energy above the reserve into miles of assist left, so you always know what you have earned.
- **No wall port.** There is no charge connector on the bike. The pack ships about half charged; after that, only your legs, regen and the sun fill it. LiFePO4 is happy sitting part-charged, so long dark stretches do no harm.




Same screen in ride mode: speed, assist budget, reserve line on the battery bar

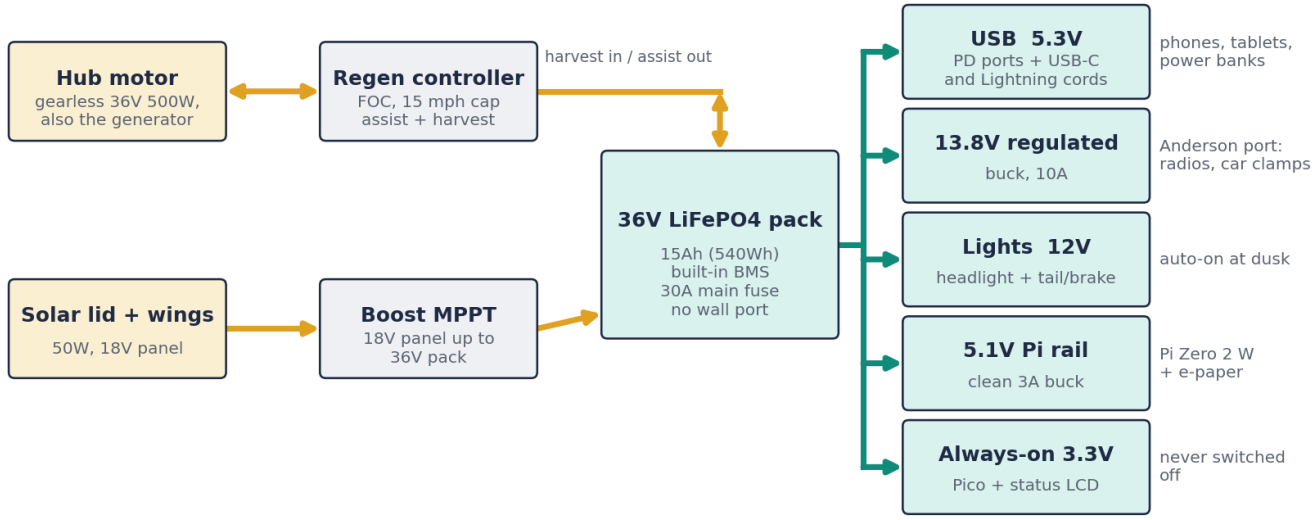
*Ride view of the status screen. The tick on the battery bar marks the 25% reserve.*

**Legal note.** With pedal-assist only, a 500W motor and a 15 mph cap, this should fit Michigan's Class 1 e-bike rules (the federal and Michigan limit is 750W). Rules vary by city and trail, so check local ordinances before riding paths.

# 4. Power system

## Power path

Only two ways in: your legs and the sun. No wall-charge port. Double arrows run both ways.

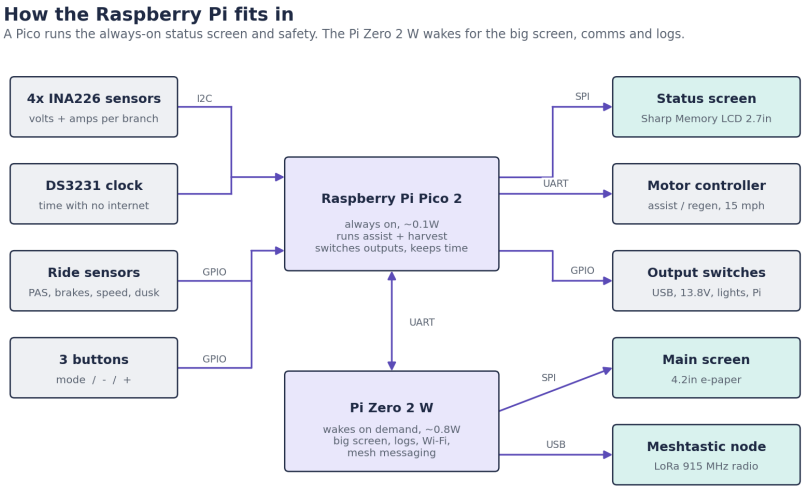



*Two ways in, one battery, five switchable outputs.*

## Key design choices

- **36V pack instead of 12V.** E-assist needs a pack that matches the motor. A 12-cell LiFePO4 pack at 36V nominal is the safe, long-life choice; every lower voltage is stepped down from it.
- **One controller does assist and harvest.** A field-oriented (FOC) controller with proportional regen, such as Grin's Phaserunner, drives the hub for assist and turns it into a generator for harvest, braking and station mode. The Pico sets the regen current over a serial link, and that current is the pedaling resistance.
- **Gearless hub only.** Geared hubs have an internal freewheel and cannot generate or regen.
- **Boost MPPT for solar.** The 18V panel sits below pack voltage, so the solar controller has to step up to 36-43V.
- **Battery-full protection.** When the pack is full the controller stops regen and the screen says so; in station mode a small dump resistor keeps the pedal feel so the generator voltage can never spike.
- **Separate low-voltage rails.** 13.8V for radios and car batteries, 12V for lights, a clean 5.1V for the Pi (its limit is about 5.25V), 5.3V for USB, and an always-on 3.3V for the Pico and status screen.
- **Fusing.** 30A main fuse within 7 inches of the battery terminal, plus a fuse on every branch.

# 5. How the Raspberry Pi integrates




## Why two computers

A Pi Zero 2 W draws 0.5-1W idling and takes time to boot. You want speed, battery and assist status the instant you glance down, and you never want a crashed Linux system in charge of a motor. So the tiny Pico (also a Raspberry Pi product) is always on: it reads every sensor, runs the status screen, enforces the 15 mph cap and reserve, switches the outputs, and powers the Pi up or down. The Pi asks the Pico for data over a serial link and takes 5.1V power on pins 2 and 6.

## Pin map

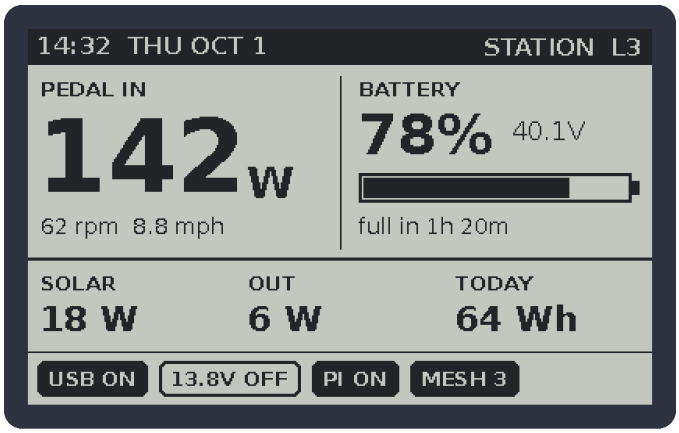
| Pico 2 pin | Connects to | Purpose |
|---|---|---|
| **GP0 / GP1** | Pi GPIO15 / GPIO14 | UART0 link to the Pi (TX / RX) |
| **GP4 / GP5** | INA226 x4, DS3231 | I2C bus: sensors at 0x40, 0x41, 0x44, 0x45; clock at 0x68 |
| **GP8 / GP9** | Motor controller | UART1: assist level, regen current, speed limit, speed readback |
| **GP10 / 11 / 12 / 27** | Load switches | USB, 13.8V, Pi rail and lights enable |
| **GP13 / GP14 / GP15** | Buttons | Mode, minus, plus |
| **GP16 / GP18 / GP19 / GP20** | Sharp LCD | CS, SCK, MOSI, DISP |
| **GP21** | Screen front-light | PWM for night reading |
| **GP22 / GP28** | PAS + brake / light sensor | Brake cutoff watch; GP28 (ADC) turns lights on at dusk |
| **GP26** | Pi GPIO26 | Pi signals 'safe to power off' |

| Pi Zero 2 W pin | Connects to | Purpose |
|---|---|---|
| **Pin 8 / 10 (GPIO14 / 15)** | Pico GP1 / GP0 | UART link (enable_uart=1, dtoverlay=disable-bt) |
| **Pin 19 / 23 / 24 (SPI0)** | E-paper | MOSI, SCLK, CE0 |
| **Pin 22 / 11 / 18** | E-paper | DC (GPIO25), RST (GPIO17), BUSY (GPIO24) |
| **Pin 37 (GPIO26)** | Pico GP26 | Shutdown-complete signal |
| **USB OTG port** | Meshtastic node | Serial to LoRa radio (node keeps relaying if Pi is off) |

# 6. The two screens

Both screens are reflective: they get clearer in sunlight and neither has a backlight. Tiny front-light LEDs cover night use, and the headlight's auto-on sensor turns them on together.

## Status screen (always on)

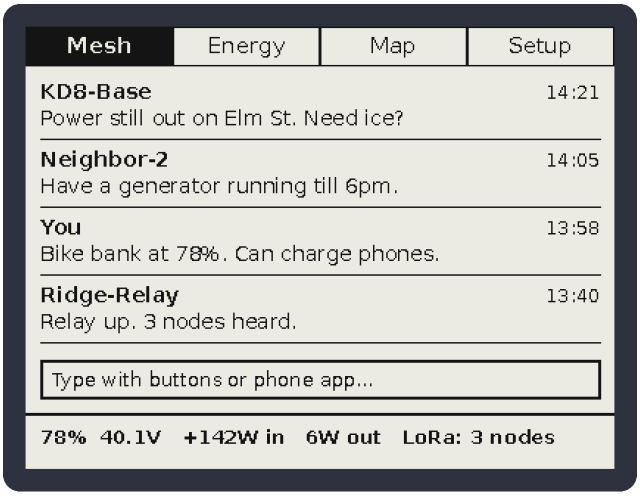



Station mode: 2.7in Sharp Memory LCD, 400 x 240, no backlight

*Top tube mount. In station mode it shows pedal watts; out riding it switches to speed and assist left (see section 3).*

- Time and date from the battery-backed clock, plus current mode and level.
- Pedal watts (station) or speed (riding), cadence, and motor power in or out.
- Battery %, voltage, reserve marker, and time to full or assist miles left.
- Solar watts, energy made today, and which outputs and lights are on.

## Main Pi screen (on demand)



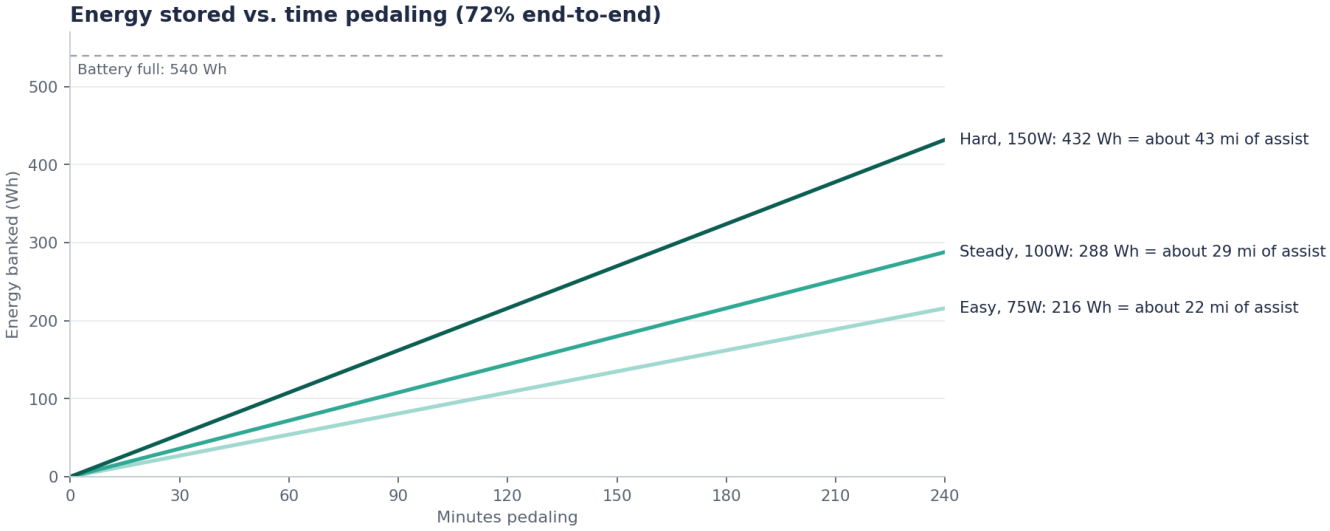

Main Pi screen: 4.2in e-paper, 400 x 300, no backlight

*Handlebar mount. E-paper holds its image with zero power, so the Pi can refresh it and go back to sleep.*

- **Mesh:** off-grid neighborhood text messages over LoRa.
- **Energy:** daily and lifetime Wh made, miles assisted, solar vs. pedal split.
- **Map:** offline neighborhood map with mesh node positions.
- **Setup:** assist and harvest levels, reserve %, output limits, node name.

# 7. Realistic energy budget

From your legs to the battery is about 72% efficient (generator about 82%, controller about 92%, battery about 95%). A fit adult can hold 100W for an hour; trained riders hold 150W or more. Assisted riding at 15 mph while you pedal too uses roughly 10 Wh per mile on flat ground, more on hills.




*Filling the whole 540 Wh pack takes about 7.5 hours at 100W, so think of pedaling as topping up, with solar doing the steady work.*

## What one hour at 100W (about 72 Wh) buys you

| Use | Approx. draw | What you get |
|---|---|---|
| **E-assist at up to 15 mph** | about 10 Wh per mile | about 7 miles of assist |
| **Smartphone full charge** | about 15 Wh incl. losses | about 5 charges |
| **Headlight + tail light** | about 6W | about 12 hours |
| **Pi + both screens + mesh node** | about 1.2W average | about 60 hours |
| **Handheld radio battery packs** | about 7 Wh each | about 10 packs |
| **Car battery top-up at 13.8V** | about 5 Ah delivered | helps a weak battery; not a cold-start replacement |

## Solar in Michigan

The 50W lid collects roughly 150 Wh on a clear summer day (about 15 miles of assist) and roughly 50-70 Wh on a clear winter day. Clouds cut that sharply. Parked in the sun all week, the solar alone keeps a commuter's assist going in summer.

# 8. Parts list and rough cost

Prices are approximate US retail as of October 2026 and move often; check before buying. Reusing a bike, trainer or battery you already own cuts the total quickly.

| Part | Low | High |
|---|---|---|
| Donor bike: used hybrid/commuter with rack mounts and disc brakes | $150 | $300 |
| Gearless 36V 500W rear hub wheel (motor + generator) | $250 | $330 |
| Regen-capable FOC motor controller (budget KT sine-wave to Grin Phaserunner) | $90 | $300 |
| PAS sensor + brake levers with motor cutoff | $30 | $50 |
| 36V LiFePO4 15Ah battery with BMS (no charger needed) | $140 | $260 |
| 50W solar panel for trunk lid | $60 | $90 |
| Optional fold-out wing panels (2 x 20W) | $0 | $60 |
| Boost MPPT solar controller (18V panel to 36V pack) | $40 | $150 |
| 13.8V 10A regulated buck converter (36V in) | $20 | $35 |
| USB-C PD + USB-A charging module | $15 | $25 |
| Retractable USB-C and Lightning cords + handlebar pod | $15 | $25 |
| Headlight (36V e-bike type) + tail/brake light | $35 | $70 |
| 5.1V Pi buck + 12V lights buck | $12 | $20 |
| Fuses, breaker, Anderson Powerpoles, wire, car clamp lead, ideal-diode module | $40 | $60 |
| Raspberry Pi Zero 2 W + microSD | $25 | $30 |
| Raspberry Pi Pico 2 | $5 | $7 |
| 4.2in e-paper display (SPI) | $30 | $40 |
| 2.7in Sharp Memory LCD breakout (Adafruit 4694) | $45 | $56 |
| 4x INA226 sensor boards + DS3231 clock + light sensor | $25 | $35 |
| Meshtastic LoRa node (Heltec V3 or RAK) + 915 MHz antenna | $40 | $60 |
| Buttons, MOSFET load switches, dump resistor, perfboard | $25 | $35 |
| Weatherproof trunk box + heavy-duty rear rack | $60 | $100 |
| Frame-triangle battery mount | $25 | $40 |
| Used wheel-on trainer (roller + flywheel) + front riser block | $50 | $120 |
| Torque arm, mounts, 3D-printed screen housings | $20 | $35 |
| Estimated total | $1,247 | $2,333 |

*Added weight is roughly 15-17 kg (hub about 6 kg, battery about 6 kg, trunk and panel about 3 kg, electronics and lights about 1.5 kg).*

# 9. Build plan

| Phase | What you build | Proves |
|---|---|---|
| **1. Bench test** | Hub wheel in a stand, controller, 36V pack, one INA226 and the Pico on a desk. | Assist, regen current control, battery-full behavior. |
| **2. Status screen** | Sharp LCD, clock and buttons on the Pico; firmware for modes, 15 mph cap and reserve. | Live readings, mode switching, safety limits. |
| **3. Station mode** | Hub on the bike, trainer and riser block adapted, fusing and dump resistor. | Real pedal feel, sustained watts, converter heat. |
| **4. E-bike mode** | PAS sensor, brake cutoffs, lights, then short test rides in a parking lot. | Smooth assist, cap at 15 mph, cutoff on braking. |
| **5. Outputs** | USB module and cords, 13.8V rail, Anderson port, car clamp lead. | Phones, a radio and a car battery charge correctly. |
| **6. Pi, comms, solar** | Pi Zero 2 W, e-paper, Meshtastic node, trunk, solar lid, weatherproofing. | Messaging, logs, sleep/wake, solar contributes. |

# 10. Safety notes

- Brake levers must cut motor power instantly; test this before every ride until you trust it.
- Use a torque arm on the hub axle; assist torque can spin an axle in a plain dropout.
- Fuse every branch; size wire for the fuse, not the expected load.
- Car clamps get reverse-polarity protection and a current limit; never connect while the engine runs.
- Keep the pack and converters out of standing water; aim for IP54 or better enclosures.
- This is a concept spec, not an engineered design. Have someone experienced with e-bike and lithium systems review it before regular use.

# 11. Open questions for you

- Wheel size and the kind of bike you have (or want to start from).
- Pedal-assist only (Class 1), or a throttle too (that makes it Class 2 and changes where you can ride)?
- How hilly is your neighborhood? That decides whether 500W and 15 mph feel like enough.
- Budget ceiling, and whether you plan to build one or make it something to sell.
- Which radios you plan to run from the 13.8V port (affects current rating).